UNIT 5 · TOPIC 5.4

Asymmetric Cryptography

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. A LightningGrid security engineer wants to send a confidential report to a colleague at another plant using PGP. Which key should the engineer use to encrypt the message?

- ○ A. The recipient's public key
- ○ B. The sender's own private key
- ○ C. The recipient's private key
- ○ D. The sender's own public key

2. A LightningGrid engineer digitally signs a classified message before transmitting it. Which key is used to create the digital signature?

- ○ A. The recipient's public key
- ○ B. The recipient's private key
- ○ C. The sender's own public key
- ○ D. The sender's own private key

3. A LightningGrid security team is selecting between 1024-bit and 3072-bit RSA keys for protecting inter-plant communications. Which statement best describes the effect of choosing the longer key?

- ○ A. A 3072-bit key is computationally more difficult to factor, providing stronger security.
- ○ B. Both key lengths provide equivalent protection against modern factoring attacks.
- ○ C. A 3072-bit key encrypts bulk data faster than a 1024-bit key.
- ○ D. A 1024-bit key is preferred because asymmetric algorithms do not benefit from longer key lengths.

4. A LightningGrid colleague encrypts a sensitive report using PGP and sends it to a security engineer. The colleague confirms the message was encrypted with the receiving engineer's public key. Which key does the receiving engineer use to read the message?

- ○ A. The sender's public key
- ○ B. The sender's private key
- ○ C. Her own private key
- ○ D. Her own public key

5. LightningGrid receives a digital certificate from a third-party Certificate Authority (CA) to authenticate its public key. A security engineer needs to verify that the

certificate is genuine and was not tampered with. Which key should the engineer use for verification?

- A. LightningGrid's own private key
- B. The CA's public key
- C. LightningGrid's own public key
- D. The CA's private key

6. NIST SP 800-131A deprecates 1024-bit RSA keys for federal use due to insufficient security strength. A LightningGrid engineer must select a replacement key length for a system that must remain secure through 2030. Which minimum RSA key length meets current NIST requirements for acceptable security strength over that period?

- A. 512-bit
- B. 1024-bit
- C. 2048-bit
- D. 1536-bit

7. LightningGrid's security architect proposes a layered communication architecture for inter-plant data transfer: asymmetric encryption handles the initial key exchange, and a symmetric session key handles bulk data encryption. Which statement best evaluates this design decision?

- A. The design is flawed; asymmetric encryption is faster and should handle all data transfer directly.
- B. The design is sound; it uses asymmetric encryption's key-distribution guarantees for the handshake while delegating bulk encryption to the computationally efficient symmetric algorithm.
- C. The design fails because a symmetric key exchanged over an asymmetric channel cannot be trusted.
- D. The design is unnecessary; modern symmetric ciphers alone can securely negotiate keys without asymmetric infrastructure.

8. A LightningGrid engineer must evaluate RSA key length recommendations for a PKI deployment protecting critical infrastructure data intended to remain confidential for 20 years. The engineer is aware that advances in quantum computing may threaten RSA within this timeframe. Which analysis best supports the key selection decision?

- A. Select 2048-bit RSA; it is the current NIST minimum and no additional analysis is required.
- B. Select 4096-bit RSA; longer classical keys provide indefinite future-proofing regardless of quantum computing advances.
- C. Abandon asymmetric cryptography entirely and use only symmetric encryption.
- D. Evaluate both current RSA key length standards and NIST post-quantum cryptography candidates, since classical key length alone does not protect against Shor's algorithm on a sufficiently powerful quantum computer.

9. LightningGrid's PKI administrator discovers that the Certificate Authority (CA) that issued all company certificates has been compromised. An attacker may have used the CA's

private key to issue fraudulent certificates. Which combination of mitigations best addresses this situation?

- ○ A. Revoke all certificates issued by the compromised CA via CRL and OCSP, remove the compromised CA from all trust stores, migrate to a new CA, and re-issue certificates only after re-verifying identities.
- ○ B. Generate new RSA key pairs for all users and no further action; new keys are unaffected by the CA compromise.
- ○ C. Extend the validity period of existing certificates to give the CA time to recover without disrupting operations.
- ○ D. Distribute the compromised CA's public key to all users so they can individually identify fraudulent certificates.

10. A LightningGrid security engineer is setting up PGP for the first time in order to exchange encrypted messages with a colleague. What is the correct first step?

- ○ A. Send a plaintext test message to the colleague to confirm connectivity.
- ○ B. Generate an asymmetric key pair.
- ○ C. Request the colleague's private key.
- ○ D. Download and decrypt the colleague's encrypted message.

11. After generating her PGP key pair, a LightningGrid engineer wants her colleague at another plant to be able to send her encrypted messages. What should she do next?

- ○ A. Send her private key to her colleague so he can encrypt messages for her.
- ○ B. Encrypt a test message using her own public key and send it to herself.
- ○ C. Ask her colleague to generate a key pair on her behalf.
- ○ D. Send her public key to her colleague.

12. A LightningGrid engineer receives a PGP-encrypted message from her colleague. Which key does she use to decrypt and read the message?

- ○ A. Her own private key
- ○ B. The colleague's private key
- ○ C. Her own public key
- ○ D. The colleague's public key

13. A LightningGrid engineer needs to send a classified document to a colleague. The document must satisfy both confidentiality (only the colleague can read it) and authenticity (the colleague can confirm it came from the engineer). Using PGP, which sequence of operations achieves both requirements?

- ○ A. Sign the document with the sender's private key, then encrypt the signed document with the recipient's public key.
- ○ B. Encrypt the document with the sender's private key, then sign it with the recipient's public key.
- ○ C. Encrypt the document with the sender's public key, then sign it with the recipient's private key.
- ○ D. Sign the document with the recipient's public key, then encrypt it with the sender's private key.

14. A LightningGrid engineer's laptop, which stores her PGP private key, is stolen. She has not yet shared a revocation certificate with anyone. What is the most appropriate immediate mitigation?

- ○ A. Generate a new key pair and silently replace her public key on the key server without notifying colleagues.
- ○ B. Continue using the old key pair until all in-flight encrypted messages are recovered and read.
- ○ C. Ask all colleagues to re-encrypt past messages they sent to her using her new key.
- ○ D. Publish a revocation certificate for the compromised key to the key server, notify all correspondents, and generate a new key pair with a new public key to distribute.

15. LightningGrid is evaluating an AI tool that automates PGP key pair generation and distributes public keys to all plants automatically. Which consideration is most critical when determining whether this tool is safe to deploy?

- ○ A. Whether the AI tool can generate key pairs faster than engineers doing so manually.
- ○ B. Whether the AI tool supports shorter key lengths to reduce processing overhead on older plant hardware.
- ○ C. Whether the AI tool has an intuitive interface that non-technical employees can use without training.
- ○ D. Whether the AI tool uses a cryptographically secure pseudorandom number generator (CSPRNG) for key generation and does not transmit or log private key material at any point.

16. LightningGrid is deploying PGP across 200 plants. The security team must choose between a centralized key generation model, where IT generates all key pairs and distributes them, and a decentralized model, where each engineer generates their own key pair locally.
Which analysis best informs this architectural decision?

- ○ A. Centralized generation is always superior because it guarantees uniform key parameters and simplifies distribution logistics.
- ○ B. Decentralized generation is inherently insecure because engineers without cryptographic expertise will generate weak key pairs.
- ○ C. Decentralized generation eliminates the risk of a central administrator becoming a single point of compromise for all private keys; the trade-off is increased operational complexity in verifying that each engineer generated conformant key parameters.
- ○ D. The choice between centralized and decentralized generation has no security implications; only the key length matters.

17. LightningGrid's CISO mandates that all cryptographic mitigations be logged for compliance with NERC CIP standards. A security engineer proposes the following logging scheme: (1) key generation events, key ID, algorithm, timestamp; (2) encryption and decryption events, message ID, key ID used, timestamp; (3) private key material, for audit purposes. Which evaluation of this proposal is most accurate?

- ○ A. The proposal is complete and should be implemented as described.

- B. Item (1) is unnecessary overhead and should be removed; items (2) and (3) are appropriate for compliance.
- C. Logging decryption events in item (2) constitutes an unauthorized interception of communications and should be removed to protect user privacy.
- D. Logging private key material in item (3) fundamentally violates the security model of asymmetric cryptography; only operational metadata, key IDs, algorithm identifiers, timestamps, and operation types, should be logged.

18. LightningGrid is piloting an AI-assisted key management system that automatically detects anomalous private key usage, such as login from an unexpected geographic location, and triggers immediate key revocation. A senior engineer raises a concern that false positives could revoke the keys of engineers legitimately traveling between plants. Which response best balances security effectiveness with operational continuity?

- A. Disable the AI system entirely to eliminate false positives and rely on manual revocation reporting.
- B. Restrict all private key usage to a fixed geographic IP range per engineer to eliminate the anomaly detection problem entirely.
- C. Accept all AI-triggered revocations without human review, since the cost of a missed true positive exceeds the cost of false positives.
- D. Configure the AI system to require human confirmation before executing revocation, maintain rapid re-issuance procedures for false positives, and log all revocation decisions, AI-triggered and human-confirmed, for post-incident review and model refinement.